

Regulatory Genomics and Epigenomic Evidence

Session 1

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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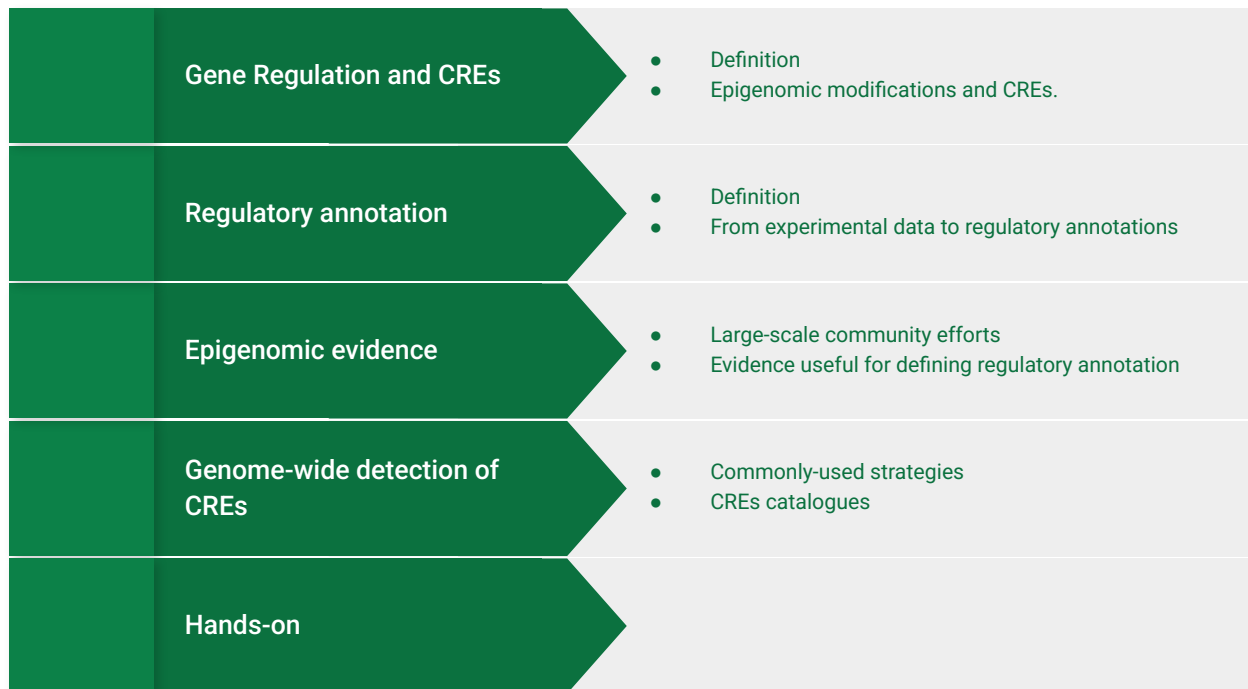
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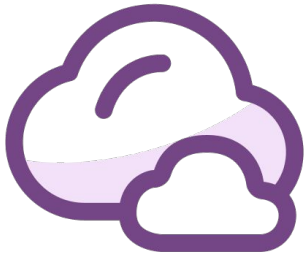
03/09/2026

Project Lead,
Ensembl Regulation
EMBL-EBI



Session map





Which are the main classes of cis-regulatory elements?

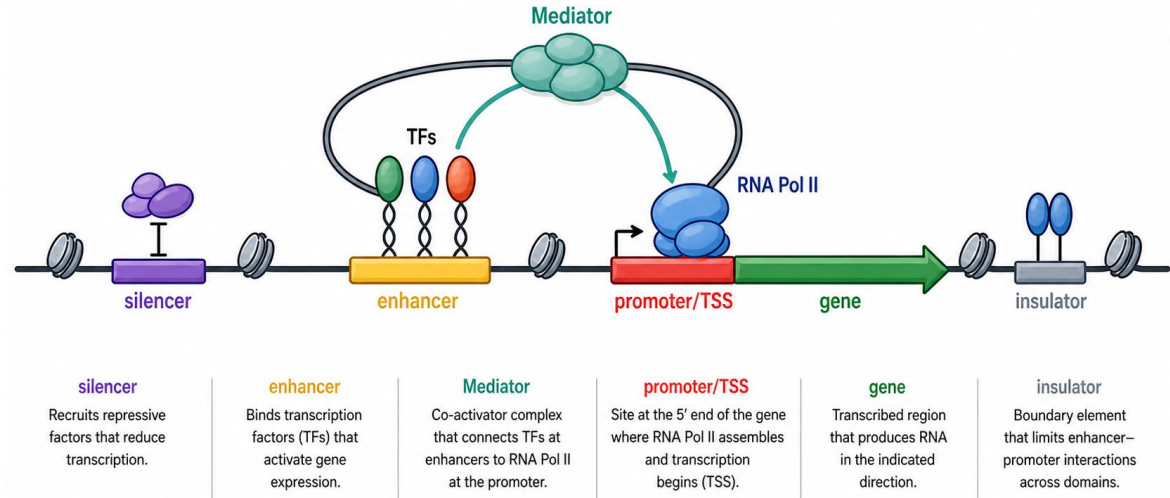


Gene regulation by CREs

- CREs are DNA sequences that regulate the transcription of nearby or distal genes on the same chromosome by recruiting transcription factors (TFs), cofactors, and chromatin-associated proteins.

- Main classes:

- Promoter
- Enhancer
- Silencer
- Insulator

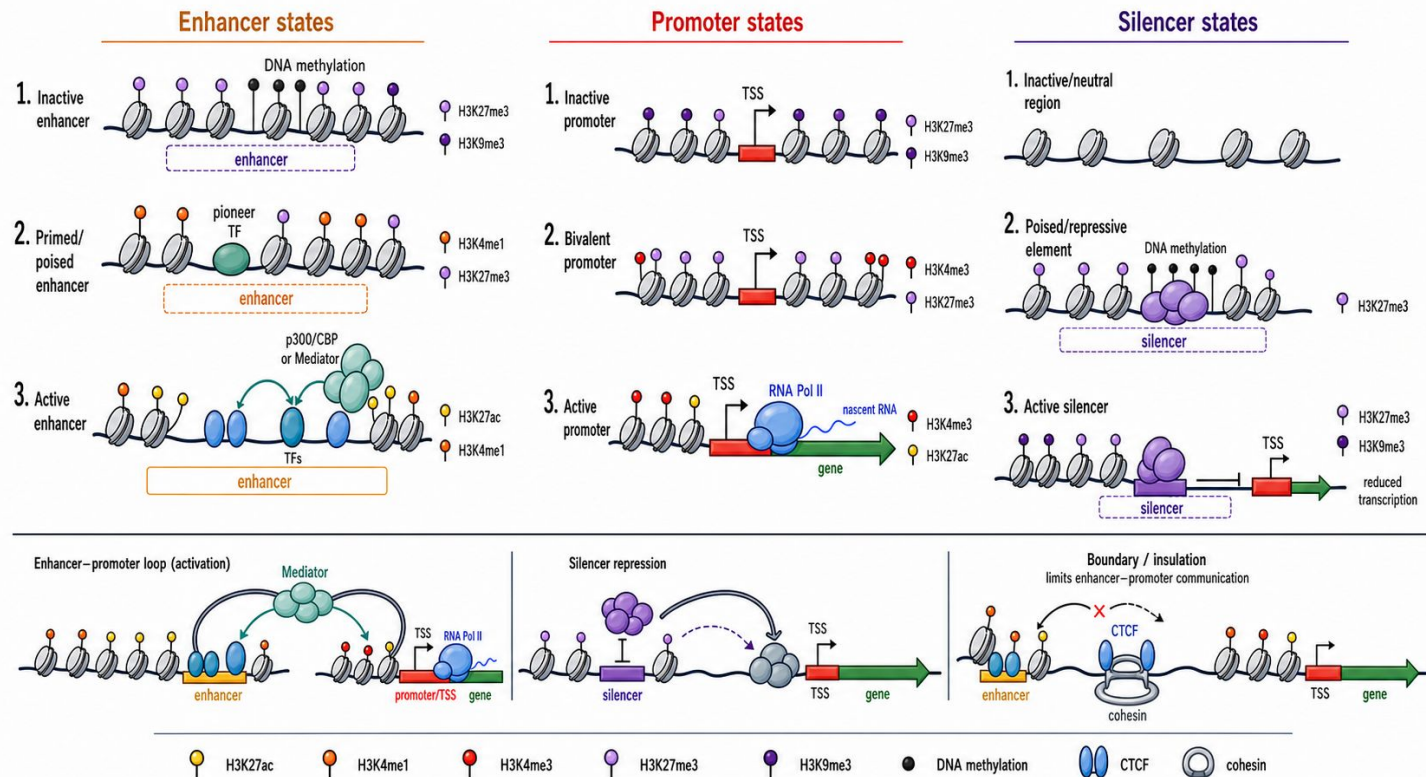


Gene regulation by CREs

Key features of CREs:

- ✓ Often located in **non-coding** regions of the genome.
- ✓ Their activity is highly cell-type, tissue, and developmental-stage **specific**.
- ✓ To be active, the DNA usually needs to be **accessible** to regulatory proteins.
- ✓ Promoters, enhancers, and silencers are often associated with characteristic chromatin features, including **histone modifications**.
- ✓ Insulators are mainly defined by the presence of architectural proteins such as CTCF and cohesin.

Epigenetic modifications and CREs



A regulatory annotation is a claim built from evidence

- ✓ Aimed to identify **potential regulatory regions** in the genome.
- ✓ Usually built by **integrating multiple evidence** layers: chromatin accessibility, histone modifications, transcription factor binding, and gene annotation.
- ✓ Based on **specific patterns** characterising the different CREs types.
- ✓ Limited to the availability of useful experimental data.
- ✓ Limited to the availability of cell-type, tissue, developmental-stage specific experiments.

Evidence useful for defining regulatory annotation

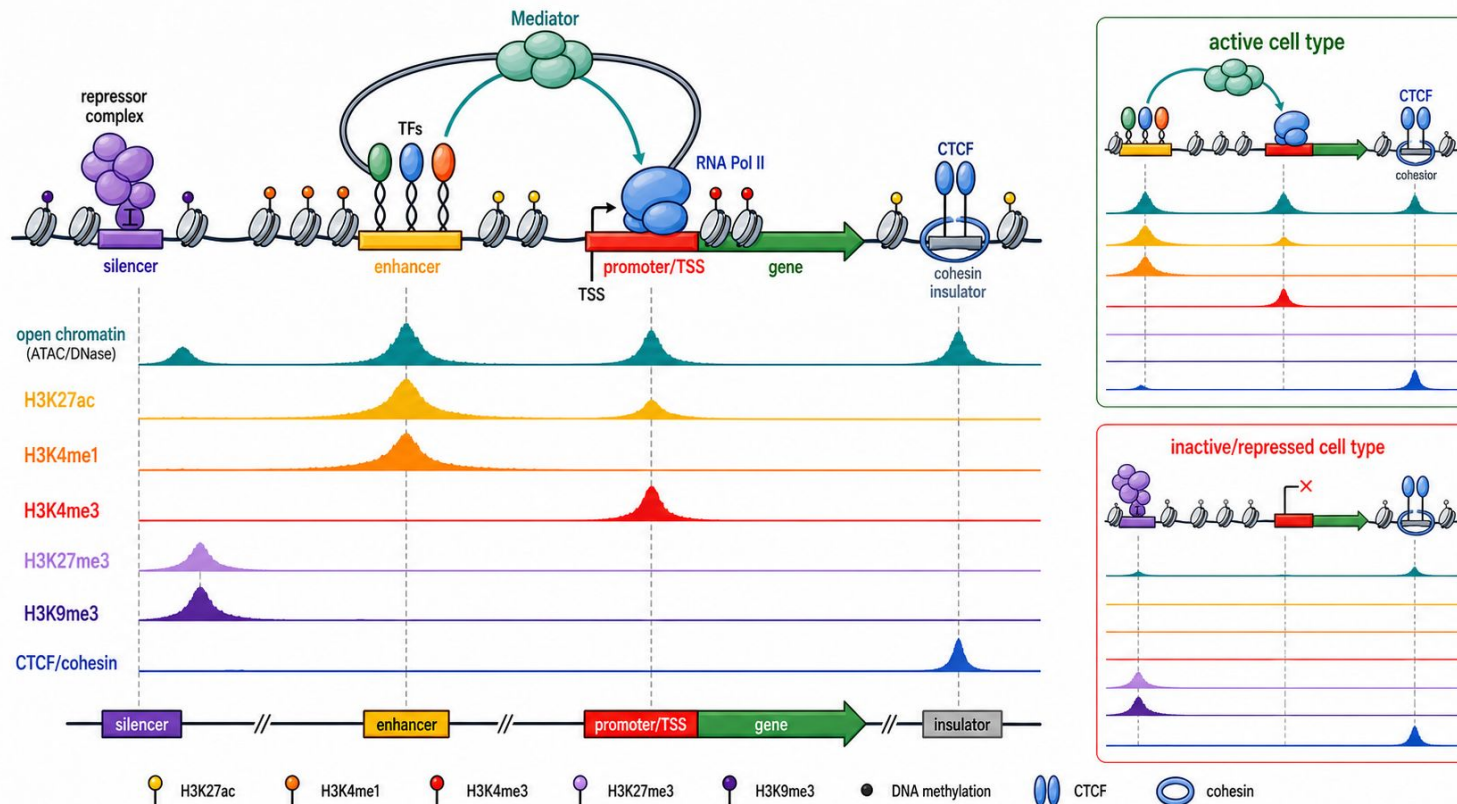
Approach	What it tells you	Advantages	Limitations
<i>Chromatin accessibility</i>	Identifies open DNA where regulatory proteins can bind	Fast genome-wide discovery of candidate CREs; useful first-pass assay	Does not distinguish enhancer, promoter, silencer or insulator by itself
<i>Histone modifications</i>	Profiles chromatin states associated with regulatory activity	Helps classify CRE types and activity states	Correlative; depends strongly on selected marks and cell type
<i>TF and cofactor binding</i>	Shows where regulatory or architectural proteins bind	Links CREs to regulatory proteins such as TFs, p300/CBP, CTCF or cohesin	Binding does not always imply functional regulation
<i>Transcription-based evidence</i>	Detects initiation or nascent transcription from promoters and enhancers	Provides evidence of active regulatory transcription	Weak or unstable enhancer RNAs can be missed
<i>Comparative genomics</i>	Finds conserved non-coding regions under evolutionary constraint	Highlights candidate functional elements across species	Misses lineage-specific or rapidly evolving CREs
<i>Functional assays</i>	Tests whether a sequence can regulate transcription	Provides direct activity evidence for tested sequences	Often outside native chromatin and genomic context

CRE annotation usually combines indirect genome-wide evidence from different sources; no single method is sufficient on its own.

From experimental data to regulatory annotation

Method	Approach	What it detects	Common use	Main limitation
ATAC-seq	Chromatin accessibility	Accessible chromatin	Candidate promoters, enhancers and other open CREs	Open chromatin alone does not define function
DNase-seq		DNase-hypersensitive accessible regions	High-confidence accessible regulatory regions	Requires more input and processing than ATAC-seq
ChIP-seq	Histone modifications / TF binding	Histone marks or protein-DNA binding	Profiles marks such as H3K27ac, H3K4me1, H3K4me3 or TF binding	Resolution and quality depend on antibody and sequencing depth
CUT&RUN		Targeted protein-DNA interactions	Lower-input profiling of histone marks or TF binding	Requires optimized antibodies and protocols
CUT&Tag		Targeted chromatin-associated proteins	Efficient profiling of histone marks and TFs	Can be sensitive to protocol and background differences

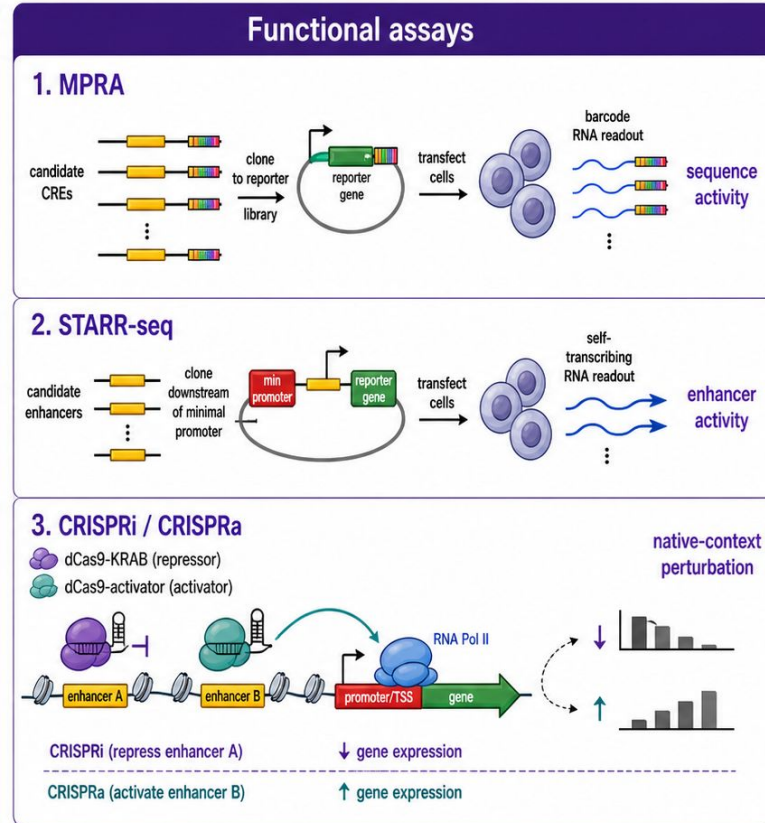
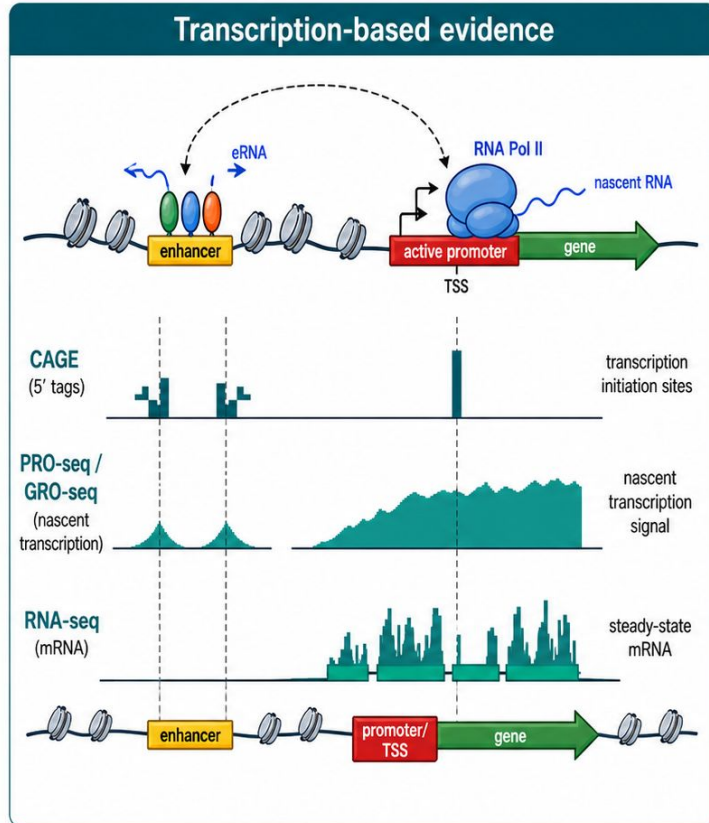
From experimental data to regulatory annotation



From experimental data to regulatory annotation

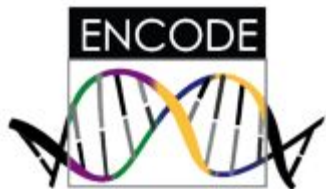
Method	Approach	What it detects	Common use	Main limitation
CAGE-seq	Transcription-based evidence	Transcription start sites	Identifies active promoters and enhancer transcription	Depends on detectable capped transcripts
GRO-seq / PRO-seq		Nascent transcription	Detects active transcription at genes and enhancers	More specialized assays and analysis
Sequence conservation	Comparative genomics	Evolutionary conserved non-coding regions	Candidate conserved CREs	Misses species-specific regulatory elements
MPRA	Functional assays	Activity of many tested sequences in parallel	High-throughput enhancer or variant testing	Removed from native genomic context
STARR-seq		Enhancer activity of cloned fragments	Genome-scale enhancer activity screening	Activity depends on assay design and cell system
CRISPRi / CRISPRa		Effect of repressing or activating candidate CREs	Links regulatory elements to target genes or phenotypes in their genomic context	Requires guide design, suitable cell model, and careful interpretation of off-target or indirect effects

From experimental data to regulatory annotation



Epigenomic evidence: Large-scale community efforts

- ✓ Community consortia that generate **standardized** epigenomic datasets.
- ✓ They cover **multiple** tissues, cell types, and species.
- ✓ Aimed to contribute to the understanding of **genome function**.



- ✓ Focused on human and mouse
- ✓ Data types: chromatin accessibility, histone marks, TF binding, DNA methylation, RNA expression, 3D genome assays.
- ✓ Outputs: peaks, genome annotations, candidate CREs.

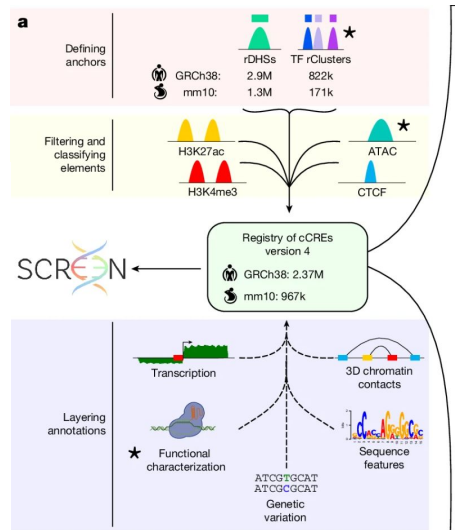
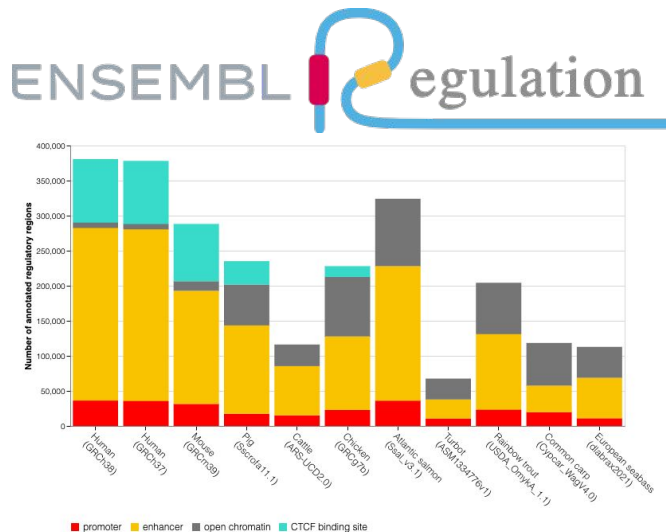


- ✓ Focused on animal genomes, especially farmed animals.
- ✓ Data types: coordinated epigenomic and transcriptomic assays across tissues and species.
- ✓ Outputs: standardized datasets, metadata, genome annotations.

Strategies for genome-wide detection of CREs

- ✓ Mainly based on strategies combining **open chromatin** and **histone/TF ChIP-seq** experiments from **large-scale efforts** involving several species.
- ✓ Focused on detecting:

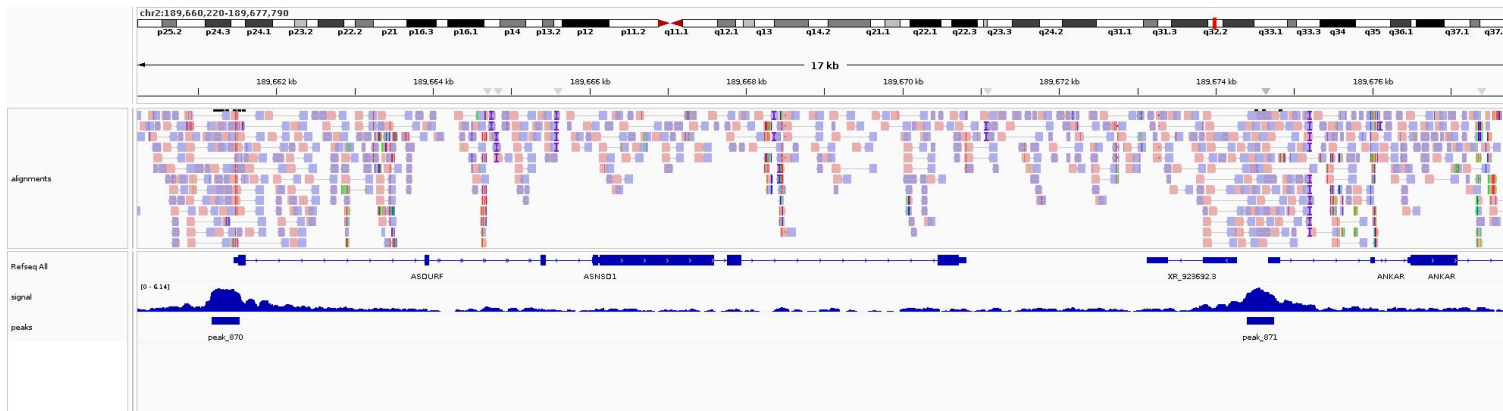
- Promoter-like sequences
- Enhancer-like sequences
- Open chromatin regions
- Transcription factor binding regions



<https://www.nature.com/articles/s41586-025-09909-9>

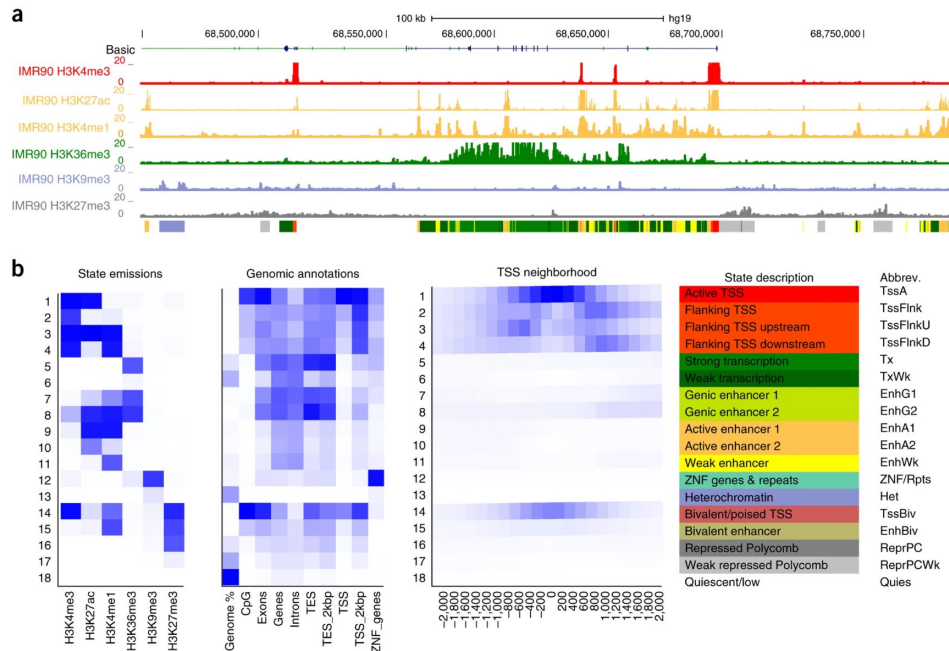
Strategies for genome-wide detection of CREs

- ✓ Require genome-wide experimental assays in several cell lines, tissues and developmental-stages
- ✓ Each assay will usually produce two tracks:
 - **Signal:** Quantitative coverage or enrichment signal
 - **Peaks:** Genomic intervals identified through statistical analysis of signal tracks



Strategies for genome-wide detection of CREs

✓ Genome segmentation by unsupervised machine learning methods



Strengths:

- Integrates several histone marks/accessibility tracks into one compact genome-wide annotation.
- Learns recurrent mark combinations de novo.
- Produces interpretable state definitions: e.g. H3K4me3 + H3K27ac-rich states suggest active promoters; H3K4me1 + H3K27ac suggests enhancers.
- Gives every genomic bin a label, making the non-coding genome easier to visualise and compare across cell types.
- Widely used and scalable.

Figure extracted from *Chromatin-state discovery and genome annotation with ChromHMM*¹.

¹Ernst, J., Kellis, M. Chromatin-state discovery and genome annotation with ChromHMM. Nat Protoc 12, 2478–2492 (2017). <https://doi.org/10.1038/nprot.2017.124>

Strategies for genome-wide detection of CREs

✓ Genome segmentation by unsupervised machine learning methods

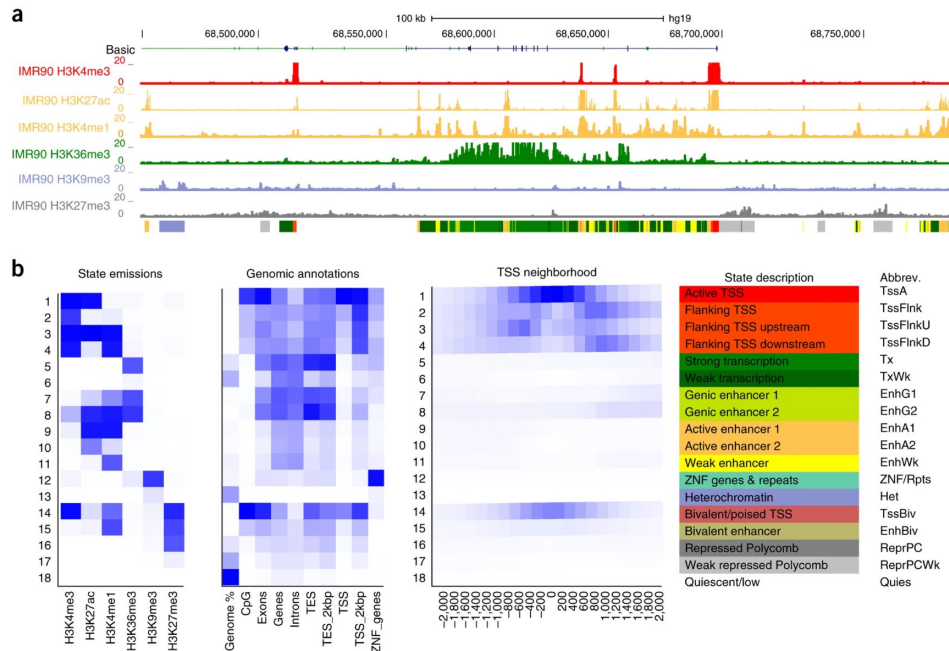


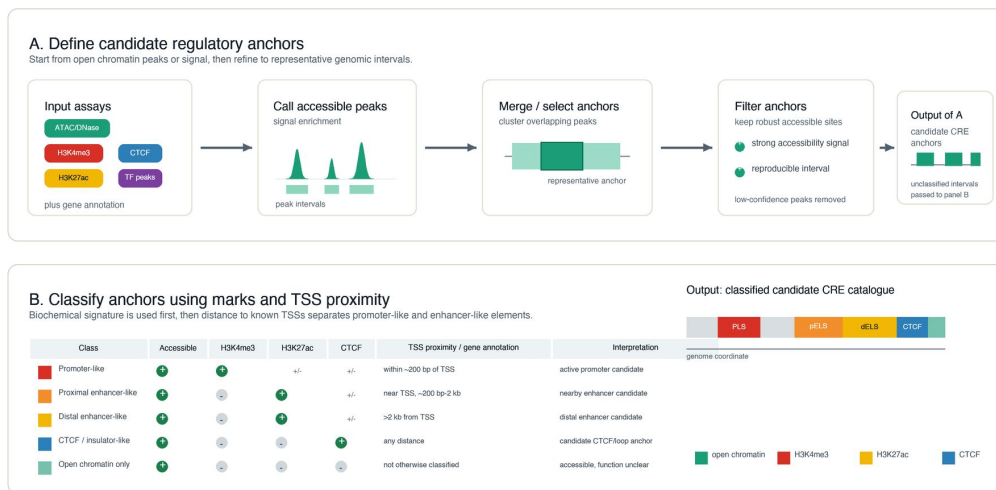
Figure extracted from *Chromatin-state discovery and genome annotation with ChromHMM*¹.

Limitations:

- State names are interpretations, **not guarantees**: an “enhancer” state is a candidate enhancer annotation, not proof that every bin with this state functions as an enhancer.
- The number of states is chosen by the user; too few states can merge biology, too many can create hard-to-interpret subclasses.
- Resolution is limited by bin size and assay resolution.
- Results depend strongly on input marks, data quality, binarisation, and missing assays.

Strategies for genome-wide detection of CREs

✓ Open chromatin-guided annotation of CREs

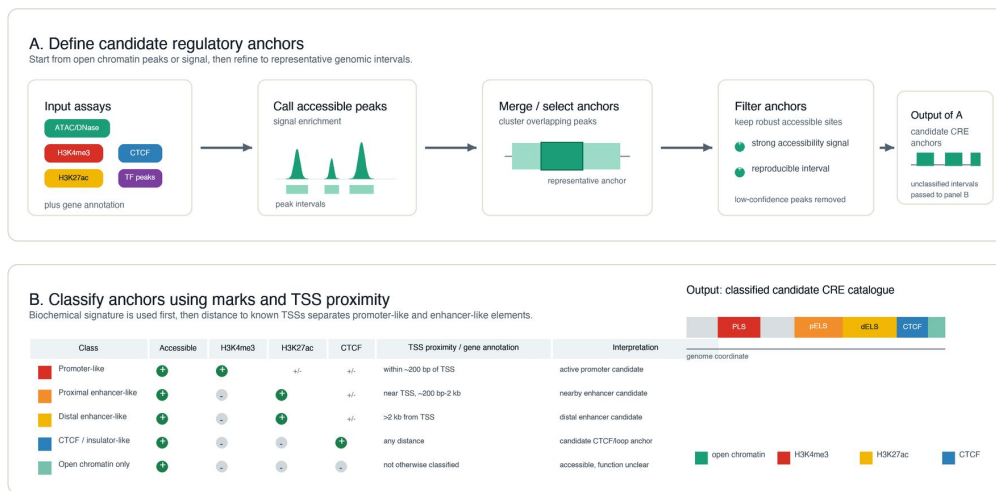


Strengths:

- Directly uses chromatin accessibility, a strong hallmark of active or poised regulatory DNA.
- Produces focal candidate regions, often easier to intersect with variants than broad chromatin states.
- More transparent than unsupervised segmentation: rules like H3K4me3 near TSS = promoter-like, H3K27ac distal = enhancer-like.
- Scales well across many tissues/cell types.
- Useful for variant interpretation because it gives a practical catalogue of candidate regulatory elements.

Strategies for genome-wide detection of CREs

✓ Open chromatin-guided annotation of CREs



Limitations:

- Open chromatin is not proof of regulatory function; it marks candidate, not validated CREs.
- Functional labels are heuristic: “enhancer-like” does not guarantee enhancer activity.
- Cell type-, tissue-, stage-, or condition-specific elements may be missed if those contexts are not represented in the open chromatin data.
- Classification depends on assay availability and quality; missing H3K27ac, H3K4me3, CTCF, or TF-binding data can limit annotation.
- Target genes, direction of effect, and causal nucleotides require extra evidence such as eQTLs, CRISPR perturbation, reporter assays, or chromatin contacts.

Questions?

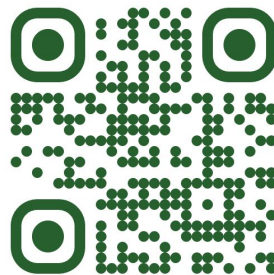


Hands-on

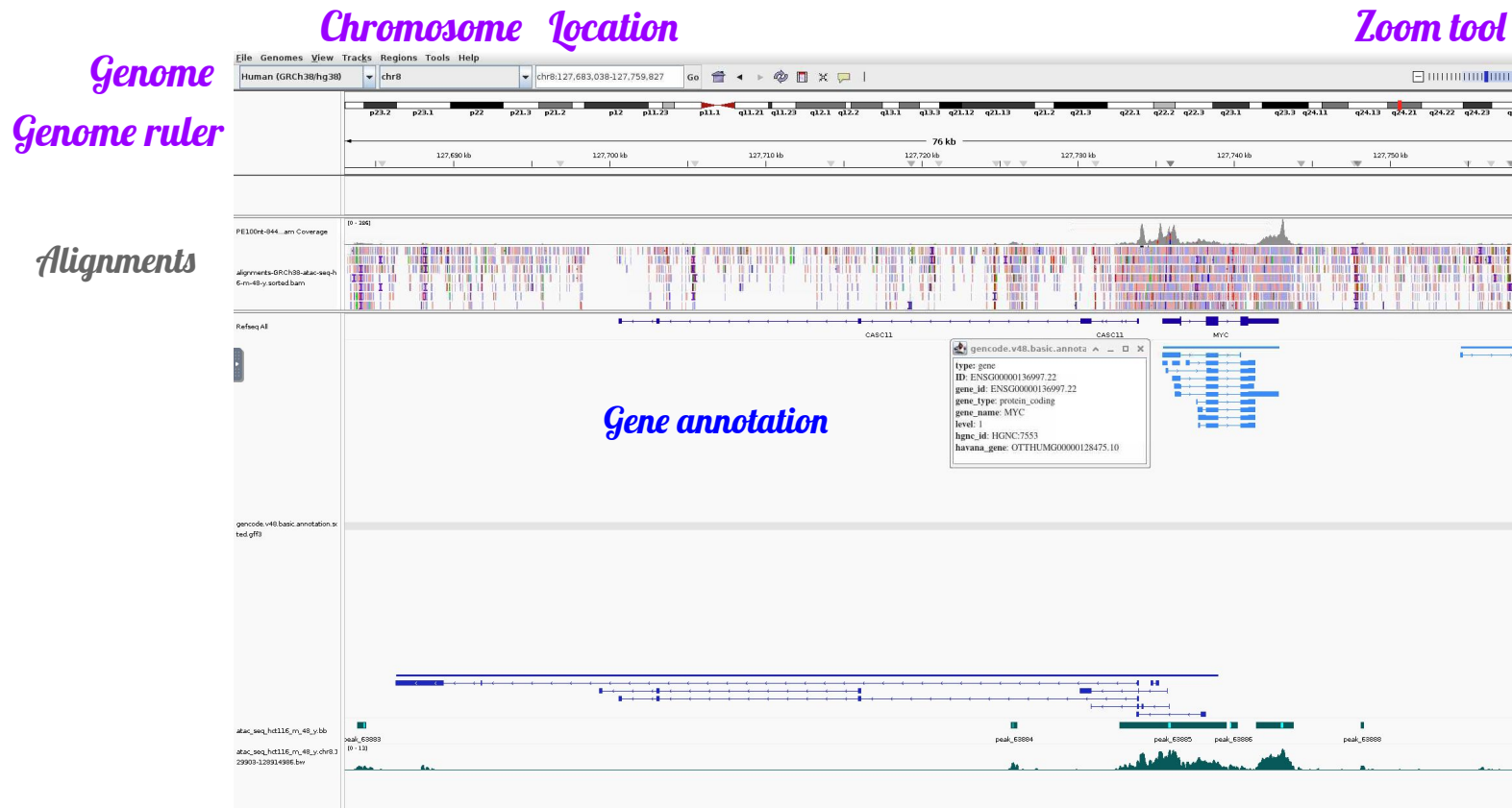


The Integrative Genomics Viewer (IGV)

- ✓ The **Integrative Genomics Viewer (IGV)** is a high-performance, easy-to-use, interactive tool for the visual exploration of genomic data. It supports flexible integration of all the common types of genomic data and metadata, investigator-generated or publicly available, loaded from local or cloud sources.
- ✓ IGV is available in multiple forms, including:
 - the original IGV - a Java desktop application,
 - IGV-Web - a web application,
 - igv.js - a JavaScript component that can be embedded in web pages (for developers)
- ✓ See <https://igv.org> for links to all forms of IGV.

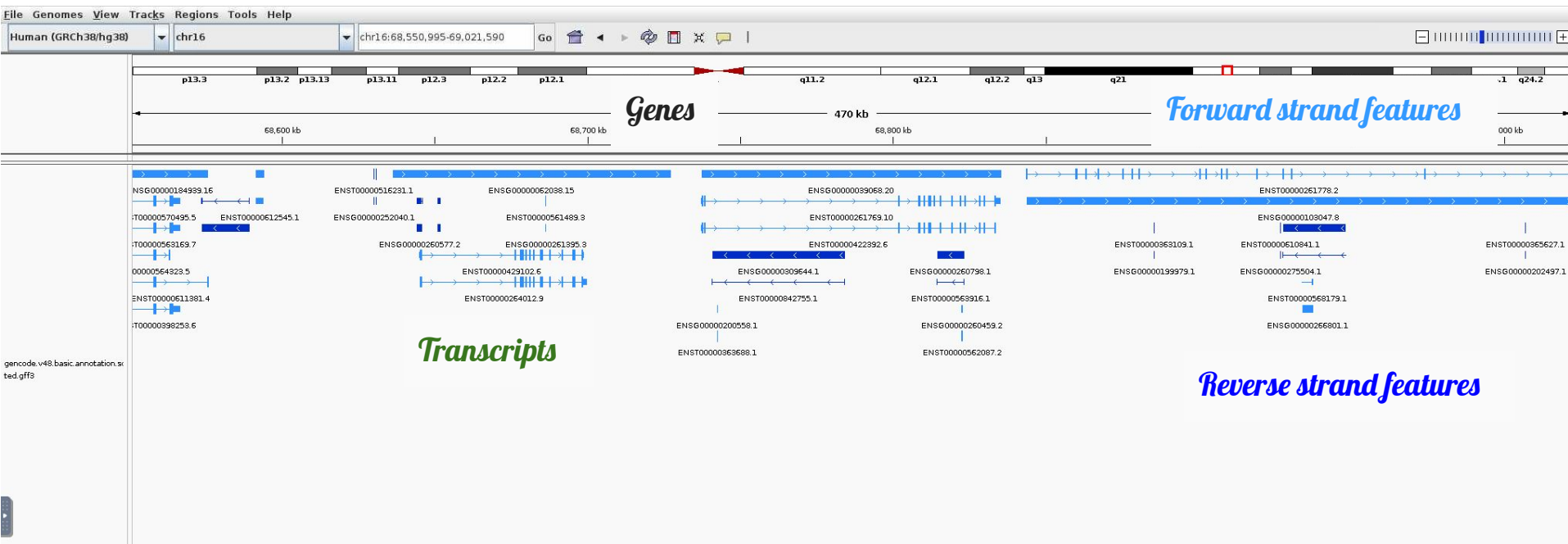


The IGV window

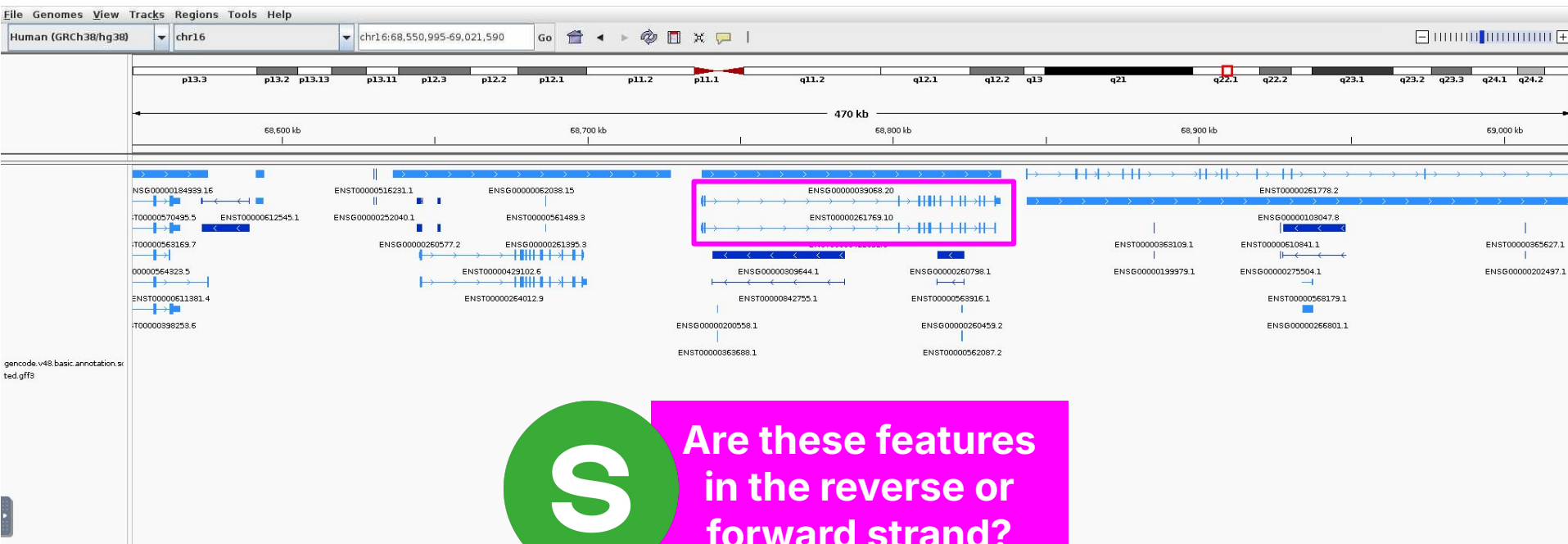


ATAC-seq signal and peaks

Understanding gene annotation tracks



Understanding gene annotation tracks



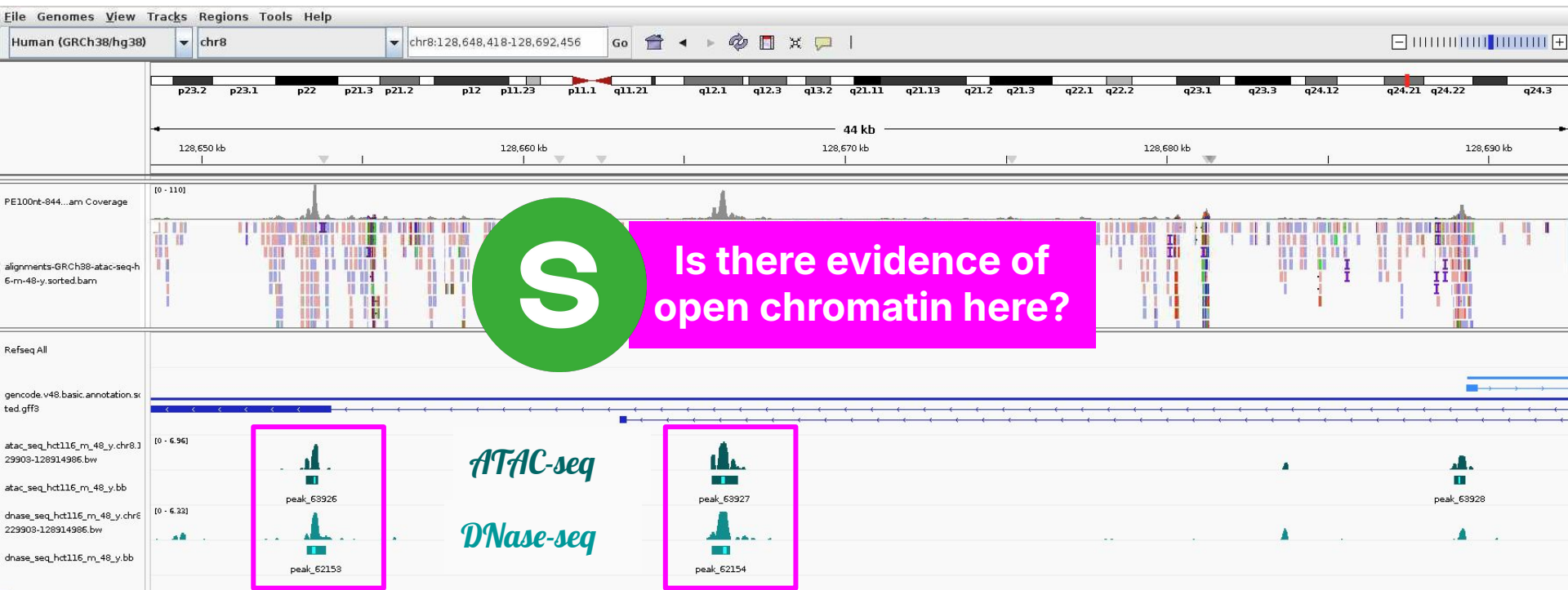
Are these features
in the reverse or
forward strand?



Are these features in the reverse or forward strand?



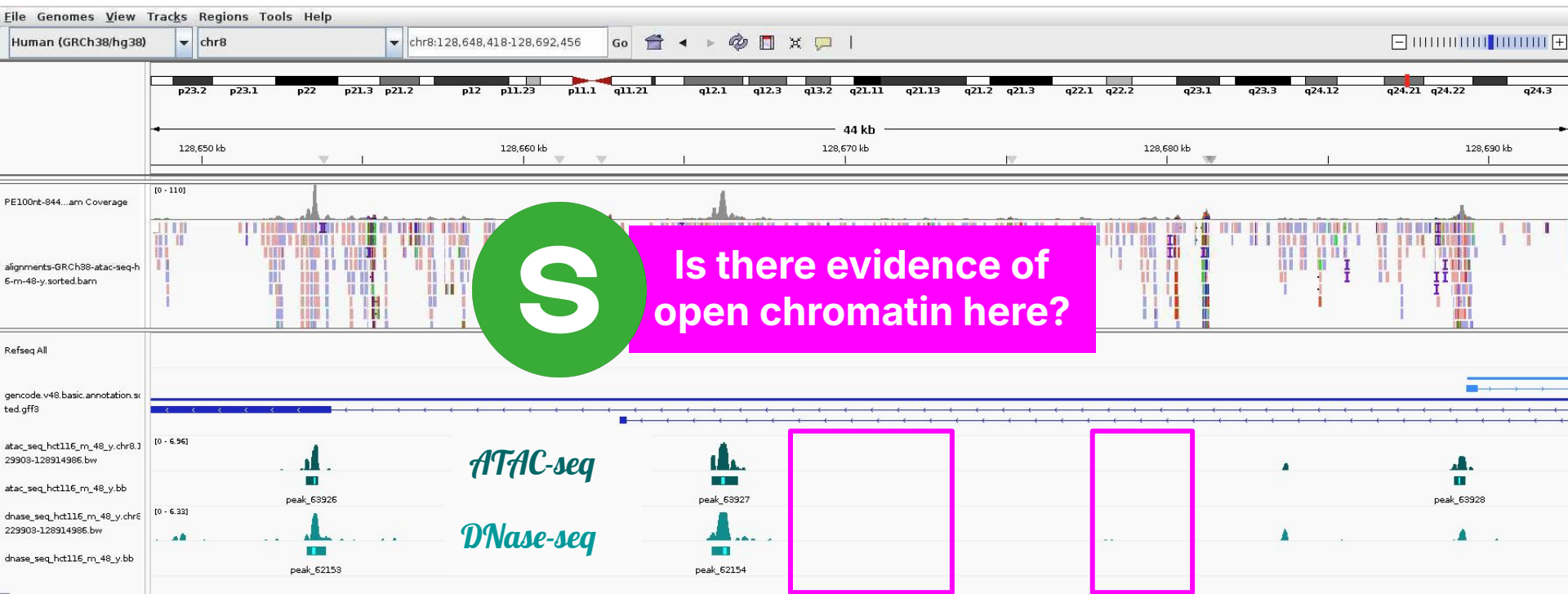
Understanding epigenomics tracks





Is there evidence of open chromatin here?

Understanding epigenomics tracks

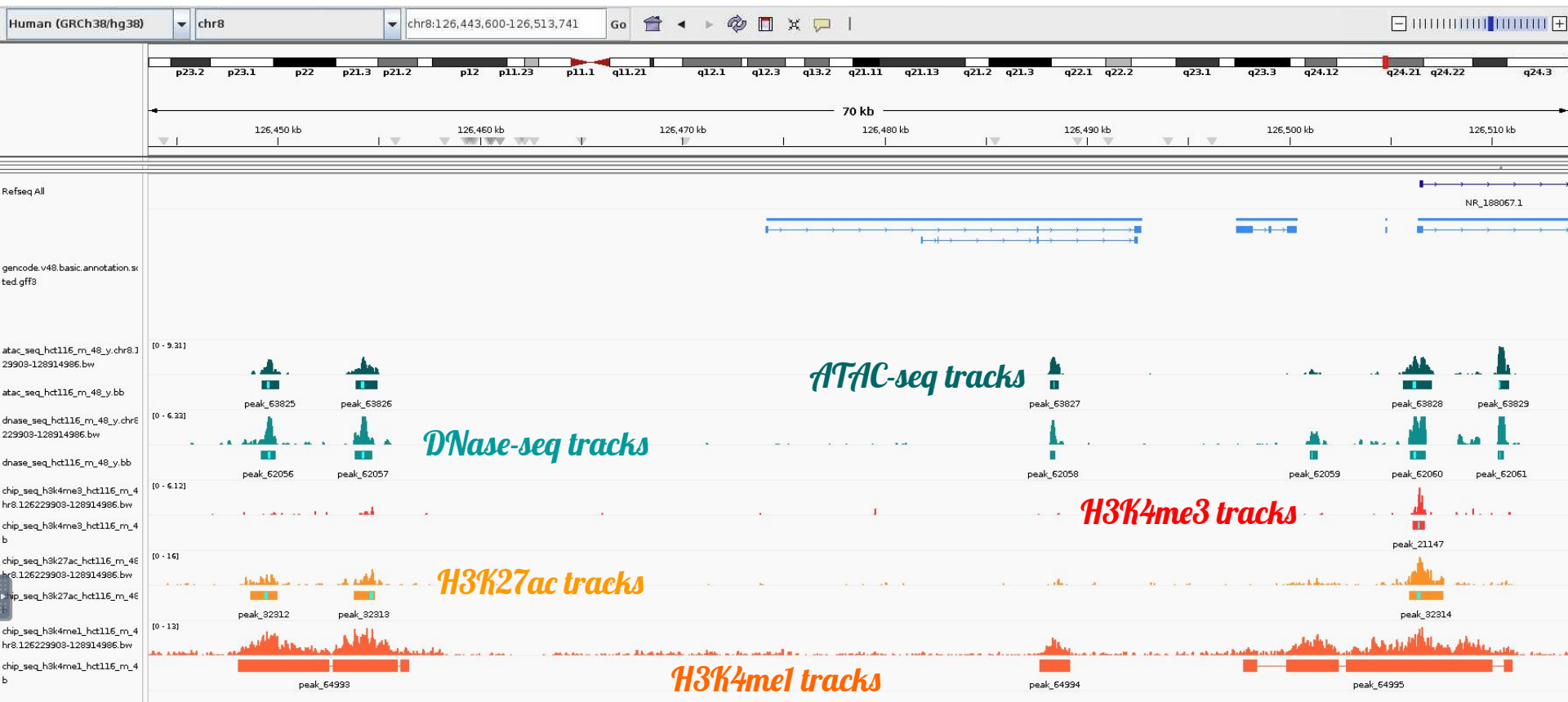




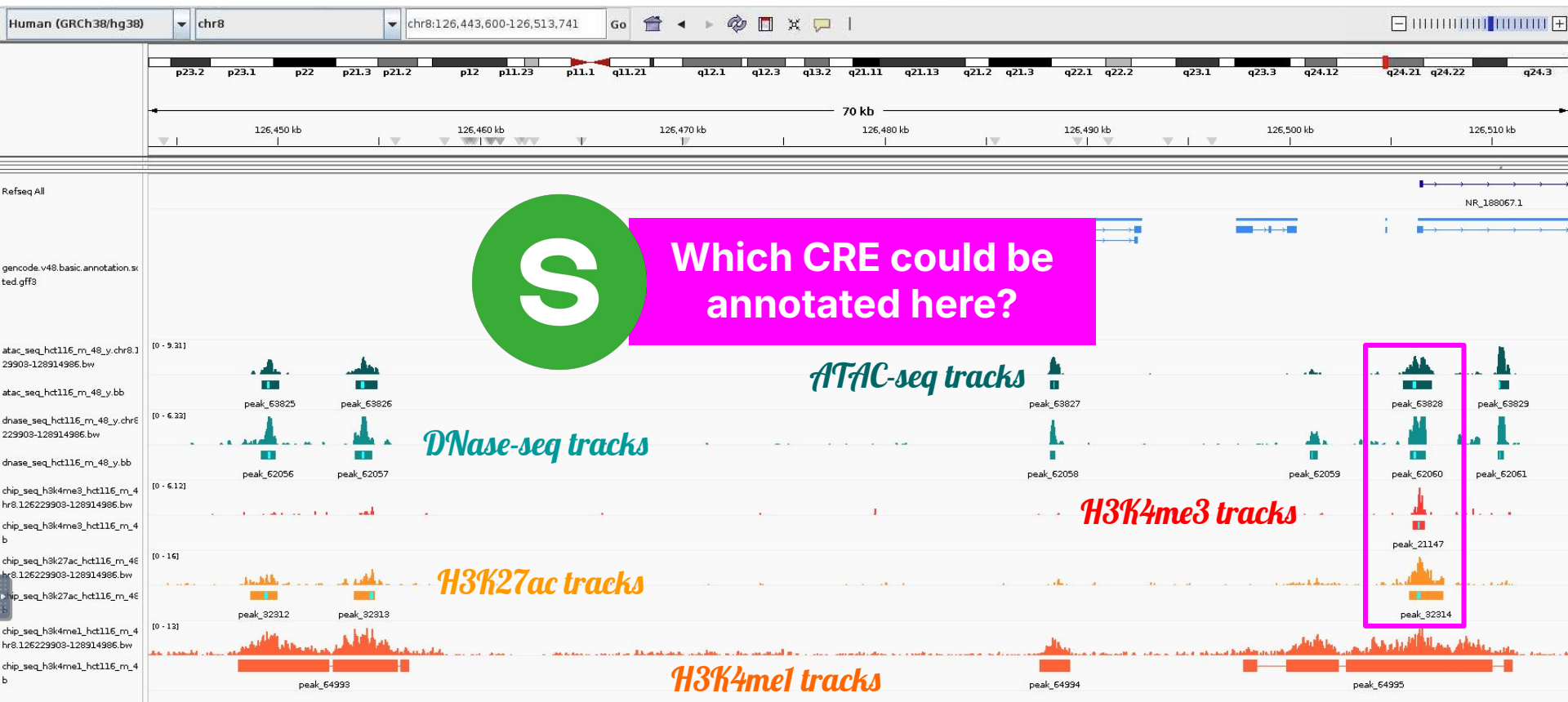
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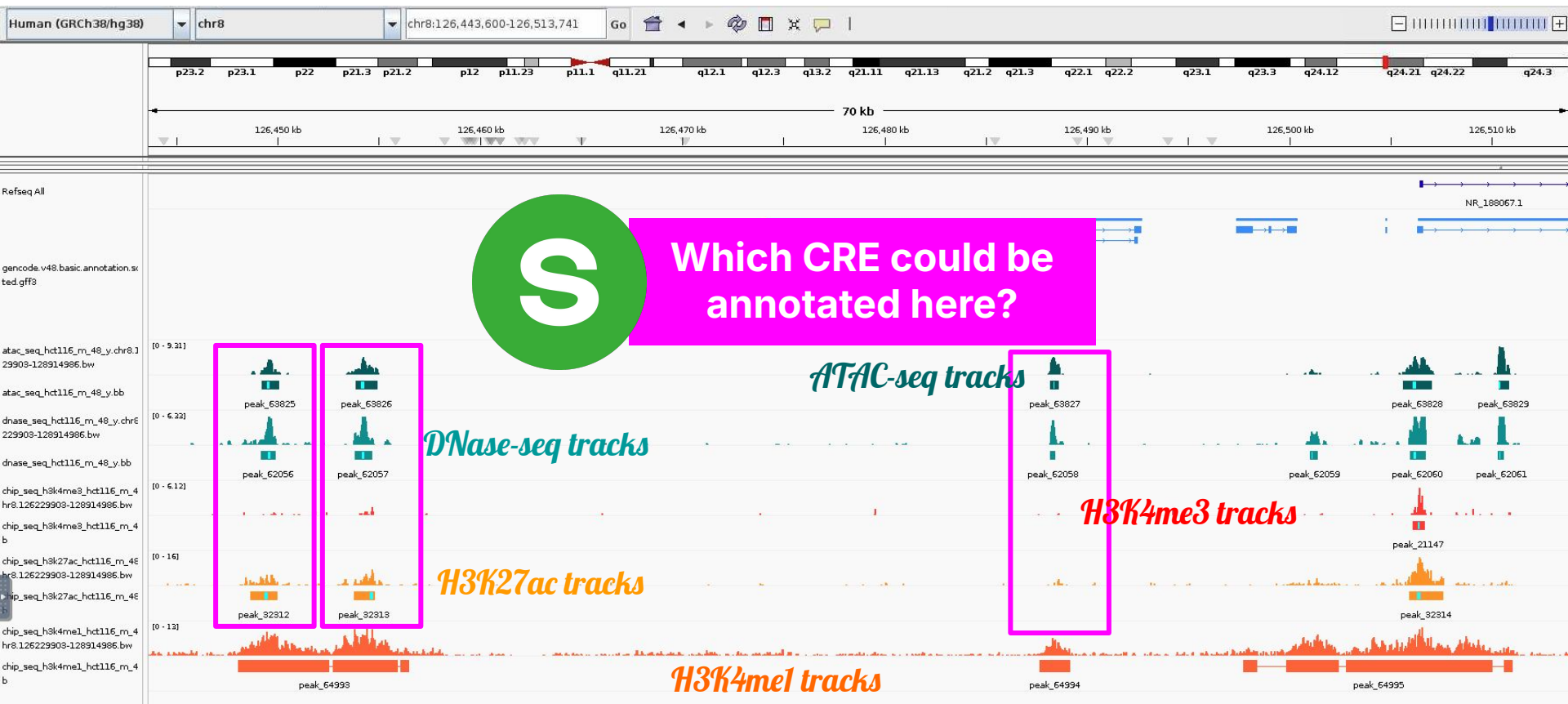




Which CRE could be annotated here?



Understanding epigenomics tracks





Which CRE could be annotated here?

